

# Towards a Reasoning Segmentation Framework for Surgical Videos

## Background and Motivation

Semantic segmentation has been extensively explored, with the evolution of neural network backbones greatly boosting the model performances.<sup>1,2</sup> However, early studies are constrained to explicit labels, which are inadequate in addressing natural language instructions that require reasoning. With LISA and V\* both demonstrating promising implicit reasoning segmentation results in the 2-D domain, our project aims to pioneer the capabilities of the video reasoning realm.<sup>3,4</sup>

**Goal:** To Achieve Video Reasoning Segmentation on Surgical Videos

## Challenges:

1. **Dataset:** Current surgical video datasets focus on tool segmentation, organ segmentation, and bleeding segmentation. Implicit reasoning examples are scarce, and manual labeling is time-consuming.
2. **Framework Set-up:**
  - **Computation cost:** The spatial-temporal information in videos needs inferencing and reasoning. This may require substantial computational resources and time if each frame is processed separately with image models such as LISA.
  - **Model Selection:** To construct a training-free agent-based framework, a zero-shot text-conditioned object detection model such as OwlV2 is needed. However, these models require fine-tuning medical content to enable medical open-vocabulary detection in surgical videos.
3. **Clinical impact:**
  - **The gradient of implicitness:** A medical surgery often requires cross-specialty cooperation, including but not limited to the physician, pathologist, radiologist, and anesthesiologist. The reasoning process should be graded into different levels: patients, medical students, resident doctors, and attending physicians.
  - **Task selection and evaluation:** The implicitness process ensures clinical storytelling and provides the possibility to compare the model's performance to that of its human counterpart (i.e., human evaluation).

### Minimum Deliverable:

1. Cultivate an **implicit reasoning segmentation dataset** composed of **cataract** surgical videos (Cateract-101, Cateract-1K, and CATERACTS).<sup>5-7</sup>
2. Construct a semi-automatic pipeline to generate an evaluation benchmark.
3. Establish a **per-framewise baseline model** using image reasoning segmentation models.
4. Implement per 3-frame or per 5-frame models, and compare the **performance** (especially Dice coefficients and processing speed) with that of the base model.
5. Implement **gradient implicitness** to evaluate the model performance in different levels of difficulties.

### Maximum Deliverable:

1. Develop an **agent-based video-reasoning segmentation framework**, including OwlV2 and V\*, and test it on the cataract dataset.
2. **Application in retinal surgical videos that meet clinical needs**
  - Dataset: TPEVGH
  - Disease: Glaucoma
  - Anatomical structure: macula lutea, optic disc, retinal detachment, etc.
3. **Human Evaluation:**
  - Testing the **model explainability** by generating reasoning chain-of-thought for open evaluation of the black-box thinking process
  - Compare segmentation **accuracy** with medical students and post-graduate resident doctors. If the result is suboptimal, we will seek more fine-tuning technology and solutions. (like BiomedCLIP and YOLO models)
4. Integrate the framework into a **web page** to provide basic real-time implicit video segmentation, including user interface optimization, interactive segmentation tools, etc.

### TO-DO list:

1. **Phase 1: Preparation:**
  - Public Datasets Labelling: Cateract-101 and CATERACTS
  - Internal Dataset: Discuss with VGHTPE, and ensure the IRB approval
2. **Phase 2: Implementation**
  - Model: OwlV2 + V\*
  - Task selection: Reasoning segmentation (medical student level)
  - Evaluation: Dice coefficient
3. **Phase 3: Evaluation and Story Design**
  - Model: OwlV2 + V\*
  - Task selection: Reasoning segmentation (PGY/ resident/ attending physician level)
  - Evaluation: Dice coefficient/ Human evaluation
4. **Phase 4: Release**
  - Prepare demo/ intro webpage
  - Organize experimental data and analysis results
  - Write the draft of the paper

## **Timeline**

- **2024/08/29:** Initial concept brainstorming
- **2024/09/05:** Literature review
- **2024/09/12: Proposal Complete**
- **2024/09/19:**
- **2024/09/26:**
- **2024/10/03: Phase 1 Complete**
- **2024/10/10:**
- **2024/10/17:**
- **2024/10/24:**
- **2024/11/01: Phase 2 Complete**
- **2024/11/08:**
- **2024/11/15:**
- **2024/11/22:**
- **2024/11/29:**
- **2024/12/05:**
- **2024/12/12: Phase 3 Complete**
- **2024/12/19:**
- **2025/2: Phase 4 Complete**
- **2025/3: MICCAI submission**

## Reference

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3. Lai, X., *et al.* LISA: Reasoning Segmentation via Large Language Model. *arXiv:2308.00692* (2023).
4. Wu, P. & Xie, S. V\*: Guided Visual Search as a Core Mechanism in Multimodal LLMs. *arXiv:2312.14135* (2023).
5. Schoeffmann, K., *et al.* Cataract-101: video dataset of 101 cataract surgeries. in *Proceedings of the 9th ACM Multimedia Systems Conference* 421–425 (Association for Computing Machinery, Amsterdam, Netherlands, 2018).
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7. El-Hindy, N., *et al.* The Cataract National Dataset Electronic Multi-centre Audit of 55 567 operations: anaesthetic techniques and complications. *Eye* **23**, 50-55 (2009).